M3403 WEEK 7 2020/21

One more property of Laplace transform is that

$\lim_{s\to\infty} F(s) = 0$, where $F(s) = \mathcal{L}\{f(t)\}$

Solving Initial Value ODE's with Laplace transform

Procedure:-

- Take Laplace transform on both sides of the equation
- Use properties of Laplace transform and the initial conditions to get the equation for the Laplace transform of the solution, and then solve the equation for the Laplace transform
- Determine the inverse Laplace transform.

Examples

Solve

① $y'' - 2y' + 5y = 0$; $y(0) = 2$, $y'(0) = 4$

② $y'' + y = t^2 + 2$; $y(0) = 1$, $y'(0) = 7$

③ $y''' - y'' + y' - y = 0$; $y(0) = 1$, $y'(0) = 1$, $y''(0) = 3$

④ $y'' + 3ty' - 6y = 1$; $y(0) = 0$, $y'(0) = 0$

⑤ $y'' - y' - 2y = -8\cos t - 2\sin t$; $y(\pi/2) = 1$, $y'(\pi/2) = 0$

Soln

① $y'' - 2y' + 5y = 0$

$\Rightarrow \mathcal{L}\{y''\} - 2\mathcal{L}\{y'\} + 5\mathcal{L}\{y\} = \mathcal{L}\{0\}$

Let $\mathcal{L}\{y\} = Y(s)$ [property ⑤]

Using the property of Laplace transform

$\mathcal{L}\{y''\} = s^2 Y(s) - s\,y(0) - y'(0) = s^2 Y(s) - 2s - 4$

$$\mathcal{L}\{y'\} = sY(s) - y(0)$$
$$= sY(s) - 2$$

Substituting into the equation we get

$$s^2Y(s) - 2s - 4 - 2sY(s) + 4 + 5Y(s) = 0$$
$$\Rightarrow Y(s)(s^2 - 2s + 5) = 2s$$
$$\Rightarrow Y(s) = \frac{2s}{s^2 - 2s + 5}$$
$$= \frac{2(s-1)}{(s-1)^2 + 2^2} + \frac{2}{(s-1)^2 + 2^2}$$

$$y = \mathcal{L}^{-1}\left\{\frac{2(s-1)}{(s-1)^2+2^2} + \frac{2}{(s-1)^2+2^2}\right\}$$
$$= 2e^t\cos(2t) + e^t\sin(2t)$$

(2). $y'' + y = t^2 + 2$

$$\Rightarrow \mathcal{L}\{y''\} + \mathcal{L}\{y\} = \mathcal{L}\{t^2\} + 2\mathcal{L}\{1\}$$
$$\Rightarrow s^2Y(s) - sy(0) - y'(0) + Y(s) = \frac{2!}{s^3} + \frac{2}{s}$$
$$\Rightarrow Y(s)(s^2+1) - s - 7 = \frac{2}{s^3} + \frac{2}{s}$$
$$\Rightarrow Y(s) = \left(\frac{2}{s^3} + \frac{2}{s} + s + 7\right)\frac{1}{s^2+1}$$
$$= \frac{2 + 2s^2 + s^4 + 7s^3}{s^3(s^2+1)}$$
$$= \frac{2}{s^3} + \frac{s+7}{s^2+1}$$

$$y = \mathcal{L}^{-1}\left\{\frac{2}{s^3} + \frac{s+7}{s^2+1}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{2!}{s^3} + \frac{s}{s^2+1} + 7\frac{1}{s^2+1}\right\}$$
$$= t^2 + \cos t + 7\sin t$$

③. $y''' - y'' + y' - y = 0$

$\Rightarrow \mathcal{L}\{y'''\} - \mathcal{L}\{y''\} + \mathcal{L}\{y'\} - \mathcal{L}\{y\} = \mathcal{L}\{0\}$

$\Rightarrow s^3 Y(s) - s^2 y(0) - s y'(0) - y''(0) - s^2 Y(s) + s y(0) + y'(0) + s Y(s) - y(0) - Y(s) = 0$

$\Rightarrow Y(s)[s^3 - s^2 + s - 1] - s^2 - s - 3 + s + 1 - 1 = 0$

$\Rightarrow Y(s) = \dfrac{s^2+3}{s^3 - s^2 + s - 1}$

$= \dfrac{s^2+3}{(s^2+1)(s-1)}$

$= \dfrac{-s-1}{s^2+1} + \dfrac{2}{s-1}$

$y = \mathcal{L}^{-1}\left\{ \dfrac{-s}{s^2+1} - \dfrac{1}{s^2+1} + 2\dfrac{1}{s-1} \right\}$

$= -\cos t - \sin t + 2e^t$

④. $y'' + 3ty' - 6y = 1$

$\Rightarrow \mathcal{L}\{y''\} + 3\mathcal{L}\{ty'\} - 6\mathcal{L}\{y\} = \mathcal{L}\{1\}$

$\Rightarrow s^2 Y(s) - s y(0) - y'(0) + 3\left[-\dfrac{d}{ds}(sY(s) - y(0))\right] - 6Y(s) = \dfrac{1}{s}$

$\Rightarrow s^2 Y(s) + 3[-Y(s) - sY'(s)] - 6Y(s) = \dfrac{1}{s}$

$\Rightarrow Y'(s) - Y(s)\dfrac{(s^2-9)}{3s} = -\dfrac{1}{3s^2}$

Which is a first order linear equation.
The integrating factor is

$$u = \exp\left(\int -\left(\frac{s^2-9}{3s}\right) ds\right)$$

$$= \exp\left(-\int \left(\frac{s}{3} - \frac{3}{s}\right) ds\right)$$

$$= \exp\left(-\frac{s^2}{6} + 3\ln s\right)$$

$$= s^3 e^{-s^2/6}$$

So,

$$\frac{d}{ds}\left(Y(s)\, s^3 e^{-s^2/6}\right) = -\frac{s}{3} e^{-s^2/6}$$

$$\Rightarrow Y(s)\, s^3 e^{-s^2/6} = e^{-s^2/6} + C$$

$$\Rightarrow Y(s) = \frac{1}{s^3} + \frac{C e^{s^2/6}}{s^3}$$

Since $\lim_{s\to\infty} Y(s) = 0$, this can only be possible if $C = 0$, thus

$$Y(s) = \frac{1}{s^3}$$

$$\Rightarrow y = \mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{2}\left(\frac{2!}{s^3}\right)\right\}$$

$$= \frac{t^2}{2} \;//$$

⑤ $y'' - y' - 2y = -8\cos t - 2\sin t \;;\; y(\pi/2) = 1,\; y'(\pi/2) = 0$

Here we cannot use Laplace transform right away since the given initial conditions are not for $t=0$. To move the initial conditions to those of $t=0$, we set

$$w(t) = y(t + \pi/2)$$
$$\Rightarrow w'(t) = y'(t + \pi/2)$$
$$\Rightarrow w''(t) = y''(t + \pi/2)$$

Also $w(0) = y(0 + \pi/2) = y(\pi/2) = 1$

$$w'(0) = y'(0 + \pi/2) = y'(\pi/2) = 0$$

Replacing t by $(t + \pi/2)$ in the equation we get

$$w''(t) - w'(t) - 2w(t) = 8\sin t - 2\cos t \;;\; w(0) = 1,\; w'(0) = 0$$

$$\Rightarrow \mathcal{L}\{w''\} - \mathcal{L}\{w'\} - 2\mathcal{L}\{w\} = 8\mathcal{L}\{\sin t\} - 2\mathcal{L}\{\cos t\}$$

$$\Rightarrow s^2 Y(s) - s w(0) - w'(0) - sY(s) + w(0) - 2Y(s) = \frac{8}{s^2+1} - \frac{2s}{s^2+1}$$

$$\Rightarrow Y(s)(s^2 - s - 2) - s + 1 = \frac{8 - 2s}{s^2+1}$$

$$\Rightarrow Y(s) = \left[\frac{8-2s}{s^2+1} + s - 1\right] \frac{1}{s^2 - s - 2}$$

$$= \frac{s^3 - s^2 - s + 7}{(s^2+1)(s-2)(s+1)}$$

$$= \frac{7}{5}\left(\frac{s}{s^2+1}\right) - \frac{11}{5}\left(\frac{1}{s^2+1}\right) + \frac{3}{5}\left(\frac{1}{s-2}\right) - \left(\frac{1}{s+1}\right)$$

$$\Rightarrow w(t) = \mathcal{L}^{-1}\left\{\frac{7}{5}\left(\frac{s}{s^2+1}\right) - \frac{11}{5}\left(\frac{1}{s^2+1}\right) + \frac{3}{5}\left(\frac{1}{s-2}\right) - \frac{1}{s+1}\right\}$$

$$= \frac{7}{5}\cos t - \frac{11}{5}\sin t + \frac{3}{5}e^{2t} - e^{-t}$$

Since $w(t) = y(t+\pi/2)$, we have

$$y(t) = w(t-\pi/2)$$

Replacing t by $(t-\pi/2)$ in the solution we get

$$y(t) = \frac{7}{5}\cos(t-\pi/2) - \frac{11}{5}\sin(t-\pi/2) + \frac{3}{5}e^{2(t-\pi/2)} - e^{-(t-\pi/2)}$$

$$= \frac{7}{5}\sin t + \frac{11}{5}\cos t + \frac{3}{5}e^{2t-\pi} - e^{-(t-\pi/2)} \quad //$$

LAPLACE TRANSFORM OF PIECEWISE FUNCTIONS

The following functions are useful in studying the Laplace transform of piecewise functions

The function

$$u(t-a) = \begin{cases} 0, & 0 \le t < a \\ 1, & t \ge a \end{cases}$$

is called a unit step function

Note that $1 - u(t-a) = \begin{cases} 1, & 0 \le t < a \\ 0, & t \ge a \end{cases}$

The function

$$\Pi_{a,b}(t) = \begin{cases} 1, & a \le t \le b \\ 0, & \text{otherwise} \end{cases} \quad \text{, where } 0 < a < b$$

$$= u(t-a) - u(t-b)$$

Piecewise continuous function can be expressed in terms of step functions

Examples

Express the function in terms of unit step function

① $f(t) = \begin{cases} 2, & 0 \le t < 3 \\ -2, & t \ge 3 \end{cases}$

② $f(t) = \begin{cases} 1, & 0 < t < 1 \\ -1, & 1 < t < 2 \\ 0, & t > 2 \end{cases}$

③ $f(t) = \begin{cases} 0, & 0 < t < 1 \\ 2, & 1 < t < 2 \\ 1, & 2 < t < 3 \\ 3, & t > 3 \end{cases}$

Soln

① $f(t) = 2[1 - u(t-3)] - 2u(t-3)$

$= 2 - 2u(t-3) - 2u(t-3)$

$= 2 - 4u(t-3)$

② $f(t) = 1[1 - u(t-1)] - 1[u(t-1) - u(t-2)]$

$= 1 - u(t-1) - u(t-1) + u(t-2)$

$= 1 - 2u(t-1) + u(t-2)$

③ $f(t) = 2[u(t-0) - u(t-2)] + 1[u(t-2) - u(t-3)] + 3u(t-3)$

$= 2u(t-1) - 2u(t-2) + u(t-2) - u(t-3) + 3u(t-3)$

$= 2u(t-1) - u(t-2) + 2u(t-3)$

Using the integral definition of Laplace transform, it can be shown that

$$\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}$$

SECOND SHIFT THEOREM

If $F(s) = \mathcal{L}\{f(t)\}$ and $a > 0$, then

$$\mathcal{L}\{f(t-a)\,u(t-a)\} = e^{-as} F(s)$$

Examples

(A) Find the Laplace transform of

① $f(t) = 2u(t-1) - u(t-2) + 2u(t-3)$

② $f(t) = \begin{cases} 0, & 0 < t < 1 \\ t, & 1 < t < 2 \\ 1, & 2 < t \end{cases}$

Sol^n

① $\mathcal{L}\{f(t)\} = 2\mathcal{L}\{u(t-1)\} - \mathcal{L}\{u(t-2)\} + 2\mathcal{L}\{u(t-2)\}$

$= \frac{2e^{-s}}{s} - \frac{e^{-2s}}{s} + \frac{2e^{-3s}}{s}$

② $\mathcal{L}\{f(t)\} = \mathcal{L}\{tu(t-1) - tu(t-2) + u(t-2)\}$

$= \mathcal{L}\{tu(t-1)\} + \mathcal{L}\{(1-t)u(t-2)\}$

$= e^{-s}\mathcal{L}\{t+1\} + e^{-2s}\mathcal{L}\{1-(t+2)\}$

$= e^{-s}\left[\frac{1}{s^2} + \frac{1}{s}\right] + e^{-2s}\mathcal{L}\{-t-1\}$

$= e^{-s}\left[\frac{1}{s^2} + \frac{1}{s}\right] + e^{-2s}\left[-\frac{1}{s^2} - \frac{1}{s}\right]$ //

(B) Determine the inverse transform of

① $F(s) = \frac{e^{-2s}}{s-1}$

② $F(s) = \frac{e^{-3s}(s-5)}{(s+1)(s+2)}$

Sol^n

①. We use the second shift theorem.

$\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} = e^t$

so

$\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s-1}\right\} = e^{t-2}u(t-2)$

(2) $$\mathcal{L}^{-1}\left\{\frac{s-5}{(s+1)(s+2)}\right\} = 7\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} - 6\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}$$
$$= 7e^{-2t} - 6e^{-t}$$

so $$\mathcal{L}^{-1}\left\{\frac{e^{-3s}(s-5)}{(s+1)(s+2)}\right\} = \left[7e^{-2(t-3)} - 6e^{-(t-3)}\right]u(t-3)$$

(3) - The current $I(t)$ in an RLC series circuit is governed by the initial value problem

$$I''(t) + 2I'(t) + 2I(t) = g(t);$$
$$I(0) = 10, \quad I'(0) = 0$$

where $$g(t) = \begin{cases} 20, & 0 < t < 3\pi \\ 0, & 3\pi < t < 4\pi \\ 20, & 4\pi < t \end{cases}$$

Sol'n

$$I'' + 2I' + 2I = 20 - 20u(t-3\pi) + 20u(t-4\pi)$$
$$\Rightarrow s^2Y(s) - 10s + 2sY(s) - 20 + 2Y(s) = \frac{20}{s} - \frac{20e^{-3\pi s}}{s} + \frac{20e^{-4\pi s}}{s}$$
$$\Rightarrow Y(s)[s^2+2s+2] = 20\left[\frac{1 - e^{-3\pi s} + e^{-4\pi s}}{s}\right] + 10s + 20$$
$$\Rightarrow Y(s) = \frac{20(1 - e^{-3\pi s} + e^{-4\pi s})}{s(s^2+2s+2)} + \frac{10s+20}{s^2+2s+2}$$
$$= \frac{20e^{-4\pi s}}{s(s^2+2s+5)} - \frac{20e^{-3\pi s}}{s(s^2+2s+5)} + \frac{10s^2+20s+20}{s(s^2+2s+2)}$$
$$= 10(e^{-4\pi s} - e^{-3\pi s})\left[\frac{1}{s} - \frac{s+2}{s^2+2s+2}\right] + \frac{10}{s}$$

$$= 10\left(e^{-4\pi s} - e^{-3\pi s}\right)\left[\frac{1}{s} - \frac{s+1}{(s+1)^2+1} - \frac{1}{(s+1)^2+1}\right] + \frac{10}{s}$$

$$\Rightarrow I(t) = 10\,u(t-4\pi)\left[1 - e^{-(t-4\pi)}\left(\cos(t-4\pi) + \sin(t-4\pi)\right)\right]$$

$$- 10\,u(t-3\pi)\left[1 - e^{-(t-3\pi)}\left(\cos(t-3\pi) + \sin(t-3\pi)\right)\right]$$

$$+ 10$$

$$= 10\left[u(t-4\pi)\left[1 - e^{-(t-4\pi)}\left(\cos t + \sin t\right)\right]\right.$$

$$\left. - u(t-3\pi)\left[1 + e^{-(t-3\pi)}\left(\cos t + \sin t\right)\right] + 1\right] //$$